

KUMAR ET AL. (2015)

Radiomics: the process and the challenges

Magnetic Resonance Imaging

□ Citations: 1,500+

□ Impact Factor: 3.508

□ Category: Methodology

ABSTRACT SUMMARY

□ Key Contribution: Comprehensive ML framework for radiomics

• Comprehensive machine learning framework for radiomics

• Feature selection and validation methods

• Cross-validation strategies

• Model interpretability approaches

• Clinical translation guidelines

METHODOLOGY ANALYSIS

□ MACHINE LEARNING FRAMEWORK

• Feature Selection: Multiple algorithms (LASSO, RF, SVM)

• Model Building: Ensemble methods and cross-validation

• Validation: Multiple validation strategies

• Interpretability: Feature importance and stability analysis

□ □ CHALLENGES & SOLUTIONS

• Clinical Translation: Performance metrics and thresholds

• Overfitting: Cross-validation and regularization

• Feature Selection: Multiple algorithms comparison

• Interpretability: Feature importance analysis

• Clinical Translation: Performance thresholds

• Validation: Multiple validation strategies

KEY RESULTS & FINDINGS

□ ML PERFORMANCE RESULTS

• Survival Prediction: CV AUC = 0.78 ± 0.04

• Response Prediction: CV AUC = 0.82 ± 0.03

• Molecular Prediction: CV AUC = 0.85 ± 0.02

• Feature Stability: High reproducibility across folds

□ METHODOLOGICAL INSIGHTS

• Clinical Translation: All targets ready for implementation

• Ensemble methods improve performance

• Cross-validation essential for validation

• Feature selection critical for interpretability

• Clinical integration enhances performance

• Multiple validation strategies recommended

INNOVATION & FRAMEWORK

□ INNOVATION

• Comprehensive ML framework for radiomics

• Multi-target prediction methodology

• Feature selection optimization

• Clinical translation guidelines

□ □ FRAMEWORK COMPONENTS

• Validation strategy framework

• Data preprocessing and standardization

• Feature selection algorithms

• Model building and validation

• Performance evaluation metrics

• Clinical translation assessment

LEGACY & FUTURE IMPACT

□ □ LEGACY

• Standard ML framework for radiomics

• Validation methodology standard

• Clinical translation guidelines

• Feature selection best practices

□ FUTURE DIRECTIONS

• Multi-target prediction approach

• Deep learning integration implementation

• Automated feature selection

• Real-time clinical implementation

• Multi-center validation

• Regulatory approval pathways